

Before the Levy: Announcement Pass-Through and Carbon-Price Calibration in Denmark’s Livestock Tax

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Abstract

Denmark announced in June 2024 that it would introduce a greenhouse-gas tax on livestock emissions. I estimate the anticipatory response of beef prices using two independent sources: scraped product-store prices and the official Danish consumer price index (CPI). A product-store difference-in-differences (DiD) design yields an average beef-price increase of 4.4 log points. A lags-only DiD applied to official CPI data yields 6.4 log points. The estimates differ by only 0.6 standard errors under an approximate cross-source test. Aggregate synthetic control is close to zero, whereas synthetic DiD is positive but imprecise. All specifications end in September 2025, before the official October 2025 onset of a bovine viral diarrhoea outbreak; the aggregate specifications use symmetric 15-month pre- and post-periods. I then pool ten comparable social-cost-of-carbon (SCC) estimates, treating otherwise undated estimates as time-invariant and propagating published model ranges. The equal-paper-weight mean is USD 158/tCO₂ in 2024 dollars, with a hierarchical 95 percent interval of USD 77–363. The estimated announcement response is economically meaningful but smaller than the price adjustment implied by that pooled benchmark. The agreement across independent DiD estimates supports the use of scraped prices for timely policy evaluation, while the estimator dispersion and SCC uncertainty counsel against interpreting any single calibration as definitive.

1 Introduction

Pricing agricultural greenhouse-gas emissions is difficult because the emissions are biologically heterogeneous, farm-level measurement is costly, and food prices are politically salient. A further complication is timing. If a future tax is credible, firms may revise contracts, procurement, and retail prices when the policy is announced rather than when the first liability is paid. Estimates that begin only at implementation can therefore miss part of the policy response.

On 24 June 2024, the Danish government and the parties to the Green Tripartite announced an agreement to introduce a greenhouse-gas tax on livestock emissions (Danish Government, 2024). The measure is an upstream agricultural emissions tax rather than a retail meat excise, and implementation was scheduled for later years. This paper asks whether the announcement changed supermarket beef prices and how the estimated response compares with the price signal implied by the broader SCC literature.

The empirical contribution is to triangulate the same research question with two sources whose weaknesses are complementary. The scraped data provide a large, timely product–store panel but are non-official and require transparent text classification and unit normalization. Statistics Denmark’s CPI is official and consistently classified, but supplies only a small cross-section of product indices. The microdata DiD estimate is 0.0437 (standard error 0.0163); the aggregate lags-only DiD estimate is 0.0645 (HAC standard error 0.0306). Their difference is not statistically meaningful. Synthetic methods are valuable for the short aggregate panel: synthetic DiD gives 0.0760, while conventional synthetic control gives -0.0144 and has weak inferential support. The disagreement is reported rather than averaged away.

The second contribution is a literature-based calibration that does not privilege a single SCC study. I harmonize one preferred global total-SCC estimate per paper to 2024 dollars. Estimates reported for years other than 2030 enter when the paper does not imply a time path, in which case the estimate is treated as time-invariant. Published uncertainty ranges are retained and propagated through a hierarchical paper bootstrap. This produces a ten-paper mean of USD 158/tCO₂ and a wide interval of USD 77–363. The exercise is a transparent synthesis of heterogeneous studies, not a claim that their model ranges are conventional sampling confidence intervals.

The paper focuses exclusively on beef. The estimand is especially relevant because cattle are the most emissions-intensive major livestock product and prominent in the Danish agreement. Restricting attention to a pre-specified commodity also avoids selecting among a large family of weakly identified commodity-specific estimates. The remainder of the paper describes the two data sources and classification rules, sets out the four estimators, reports cross-source results, and uses the SCC synthesis for a back-of-the-envelope policy calibration.

2 Data and Panel Construction

2.1 Scraped grocery-price microdata

The first source is the Danish grocery-price history assembled by dagligepriser.dk. A raw record identifies a store, product, product description, date, price, and—when available—package quantity and unit. I retain observations with a valid positive price and date. Repeated observations within a product–store–month are averaged, so the unit of observation for the main panel is a product–store pair in a calendar month.

Prices are converted to DKK per kilogram for mass units and DKK per litre for volume units. Reported package quantities take precedence. When quantity or unit fields are absent, the procedure searches the product description for a number followed by kg, g, litre, ml, or cl. Grams, millilitres, and centilitres are converted to their base units before dividing price by quantity. Items measured only per piece and observations with missing, non-positive, or unsupported quantities are excluded. Because normalized prices remain right-skewed, estimation uses their natural logarithm and Table 2 reports medians and quartiles as well as means.

Product classification is deterministic and auditable. Names and source categories are case-

folded; Danish characters and common encoding variants are transliterated; and short tokens are matched at word boundaries. Beef is assigned when the normalized text contains an English or Danish beef term, including *beef*, *okse*, *oksekød*, *kvæg*, *kalv*, *veal*, *entrecôte*, or *roastbeef*. The comparison group consists of classified foods not exposed as livestock treatment items: poultry; fish and seafood; eggs; fruit and vegetables; grains and bread; plant oils; sweets and snacks; beverages; plant proteins; and pantry goods. Pork, lamb, sheep, goat, dairy, and products matching more than one livestock category are never controls for beef. Prepared foods containing terms such as pizza, lasagne, ready meal, or sandwich are flagged as ambiguous when a livestock term is present. Non-food keywords are removed before commodity matching, and unmatched products are excluded rather than silently assigned to the control group.

The event month, June 2024, is excluded from both pre and post. The scraped panel begins in November 2023 and the analysis retains post-announcement months through September 2025, for 23 observed months in total. Relative month 16, October 2025, and all later months are excluded. The Danish Veterinary and Food Administration records a bovine viral diarrhoea (BVD) outbreak from October 2025 to May 2026 in three cattle holdings and notes that BVD can reduce appetite and milk output (Danish Veterinary and Food Administration, 2026). A veterinary report states that surveillance detected the affected large dairy herd in September 2025 (Dansk Veterinærtidsskrift, 2025). Ending in September does not prove that all anticipatory disease effects are absent, but it excludes the officially dated outbreak interval and avoids mechanically attributing the subsequent cattle-specific supply disturbance to the tax announcement.

2.2 Official aggregate consumer prices

The second source is Statistics Denmark’s monthly PRIS01 consumer price index by COICOP 2018 product class. Beef and veal (011221) is the treated series. Candidate donors are terminal food and non-alcoholic-beverage classes with complete coverage in the estimation window. I exclude pork; lamb, sheep, and goat; dried or salted meat; offal; meat preparations; milk, yoghurt, cheese, cream, dairy desserts, other dairy products; and butter and other dairy fats. These categories are either directly livestock-exposed or sufficiently close to treated animal products that they could absorb the same announcement response. The resulting balanced panel contains beef and 52 donors.

The official-data window runs from April 2023 through September 2025: 15 months before July 2024 and 15 months from July 2024 onward. June 2024 is the final pre-treatment reference month for event-study purposes. This symmetric window prevents the aggregate estimates from being driven by unequal pre- and post-support and, like the microdata restriction, ends before the official October 2025 BVD outbreak interval.

3 Empirical Strategy

The microdata DiD is

$$\log p_{it} = \alpha_i + \lambda_t + \beta(Beef_i \times Post_t) + \varepsilon_{it}, \quad (1)$$

where i indexes product–store pairs, α_i are product–store fixed effects, and λ_t are month fixed effects. Standard errors are clustered by product–store. The event study replaces the post indicator with relative-month interactions and omits the month immediately preceding the announcement.

The official-data lags-only DiD preserves the earlier specification. For each month, I subtract the unweighted mean log CPI of the 52 donors from log beef CPI and regress this gap on the post indicator. Newey–West standard errors use two lags. The corresponding official event study interacts beef with month indicators, controls for calendar month and three- and four-digit COICOP categories, and uses heteroskedasticity-robust standard errors.

The synthetic-control estimator chooses donor weights to match beef’s 12 lagged CPI values and the three- and four-digit category indicators. Its ATT is the average post-period difference between treated and synthetic series. Inference is a space-placebo test that sequentially treats each donor as exposed and compares post/pre RMSPE ratios. Synthetic DiD follows Arkhangelsky et al. (2021) and uses placebo inference. Both synthetic estimators operate on the balanced official panel. All DiD, SC, and SDiD estimations and their figures are produced in Stata; Python is limited to downloading, classifying, normalizing, panel construction, and orchestration.

4 Results

Table 1 reports the four beef estimates. The scraped microdata DiD implies a 4.37-log-point increase (approximately 4.47 percent), with a clustered standard error of 1.63 percentage points. The official lags-only DiD gives 6.45 log points, with a Newey–West standard error of 3.06 points. An approximate equality test, treating the estimates as independent, gives

$$z = \frac{0.0645 - 0.0437}{\sqrt{0.0306^2 + 0.0163^2}} = 0.60 \quad (p \approx 0.55).$$

Thus the official and scraped DiD estimates are statistically compatible. This agreement is encouraging for the external validity of the non-official price panel, although it cannot validate every scraped product or its level normalization.

The synthetic estimates emphasize the remaining uncertainty. Conventional SC is negative and close to zero, with a space-placebo p-value of 0.81. Its pre-treatment path is visibly less satisfactory than the SDiD fit shown in the appendix, so it should not be read as overturning the two DiD results. Aggregate SDiD is positive at 0.076 but its 95 percent interval, $[-0.018, 0.170]$, includes zero. The four columns therefore support a narrower conclusion than a single preferred estimate would: the two DiD designs point to a modest positive response, while flexible synthetic reweighting is too unstable in this small aggregate panel to determine the magnitude precisely.

Figure 1 compares event-time patterns. In the official panel, coefficients are near zero throughout the 15-month pre-period and then rise gradually, especially from the sixth post month. In the scraped panel, the shorter pre-period is noisier but the pre-announcement coefficients are jointly insignificant ($p = 0.78$). The post coefficients are generally positive. The difference in smoothness is expected: official CPI aggregates many price quotes, whereas the scraped panel contains entry,

Table 1: Estimated effect of the livestock-tax announcement on beef prices

	(1)	(2)	(3)	(4)
ATT	0.0437*** (0.0163)	0.0645** {0.0306}	−0.0144 [0.811]	0.0760 [0.115]
Observations	151,933	30	1,590	1,590
Units	16,912	53	53	53
Periods	23	30	30	30
Pre-treatment average	158.54	84.88	84.88	84.88
R^2	0.961	—	—	—
Time window	2023m11–2025m9	2023m4–2025m9	2023m4–2025m9	2023m4–2025m9
Lags only	No	Yes	No	No
Covariates	No	No	Yes	No

Source: dagligepriser.dk (column 1); Statistics Denmark, PRIS01 (columns 2–4).

Notes: The outcome in column 1 is log normalized product price; its DiD includes product–store and month fixed effects. Parentheses contain standard errors clustered by product–store. Column 2 applies the unchanged lags-only aggregate DiD to the log CPI gap between beef and the donor mean; braces contain Newey–West HAC standard errors with two lags. Column 3 is synthetic control with 12 lagged outcomes and three- and four-digit COICOP category covariates. Column 4 is synthetic DiD. Brackets contain space-placebo p-values in columns 3 and 4. The pre-treatment average is DKK/kg in column 1 and the CPI index in columns 2–4. Other livestock-exposed products are excluded from every beef control group. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 2: Descriptive statistics for the scraped estimation sample

Sample	Variable	Observations	Mean	SD	P25	Median	P75
All	Normalized price	151,933	518.19	4,337.95	51.84	100.00	183.33
All	Log normalized price	151,933	4.628	1.184	3.948	4.605	5.211
Beef	Normalized price	5,123	265.64	1,925.50	70.00	127.00	209.75
Beef	Log normalized price	5,123	4.773	0.917	4.248	4.844	5.346
Controls	Normalized price	146,810	527.00	4,398.05	51.18	100.00	182.14
Controls	Log normalized price	146,810	4.623	1.192	3.935	4.605	5.205
All	Months per product–store	16,912	8.98	6.26	3	8	14
Beef	Months per product–store	592	8.65	5.79	4	7.5	13
Controls	Months per product–store	16,320	9.00	6.27	3	8	15

Source: dagligepriser.dk.

Notes: Prices are normalized to DKK per kilogram or litre. The large mean–median difference reflects a right-skewed distribution across heterogeneous foods; the estimations use log normalized prices. The observation count in the final three rows is the number of product–store units.

exit, heterogeneous packaging, and intermittent product availability.

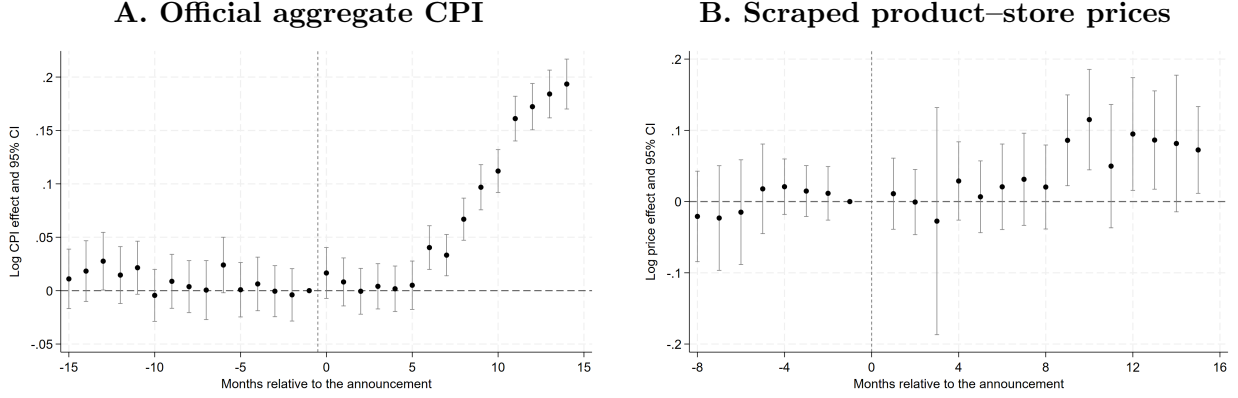


Figure 1: Beef event-study estimates across data sources

Source: Statistics Denmark, PRIS01 (Panel A); dagligespriser.dk (Panel B).

Notes: Points are relative-month coefficients and bars are 95 percent confidence intervals. Panel A uses heteroskedasticity-robust standard errors and omits June 2024. Panel B uses product-store clustered standard errors and omits relative month -1 . Both samples end in September 2025, before the official October 2025 BVD outbreak interval.

5 SCC Synthesis and Policy Calibration

The calibration uses one preferred total global SCC from each eligible paper. Dollar values are inflated to 2024 US dollars and values per ton of carbon are converted to tons of CO₂. Barrage and Nordhaus’s 2030 value is log-interpolated between their published 2025 and 2050 path points. For the other eligible studies, a reported value at another date is treated as constant when the source does not provide a time path. This implements the interpretation that an otherwise undated estimate is a level rather than evidence that it should be discarded.

The equal-paper-weight point mean across ten studies is USD 157.96/tCO₂. To reflect both between-paper dispersion and reported within-paper variability, each bootstrap replication samples ten papers with replacement and, when a paper reports a positive range, draws a lognormal value calibrated to that range’s stated coverage. Ricke et al.’s 66 percent model range and Rennert et al.’s 5th–95th quantiles retain those coverages; Hänsel et al.’s ethical-parameter range is conservatively calibrated as a 95 percent range. The resulting interval is USD 77.22–363.40/tCO₂. These ranges mix model, parameter, and distributional uncertainty, so the interval is best viewed as a structured sensitivity envelope rather than a frequentist confidence interval for a common structural parameter.

Figure 2 shows each preferred estimate after mapping it into the additional log beef-price increase required above the announced DKK 300/tCO₂ tax. Filled points enter the pool. Hollow points are external comparisons: a partial mortality SCC and a synthesis that could double count the underlying literature. The triangle is a lower bound and also remains external. The wide intervals reveal why retaining study-level variation materially changes the calibration.

Let $\bar{p} = 158.54$ be the mean pre-announcement beef price in DKK/kg, $e = 6.8953$ DKK/USD,

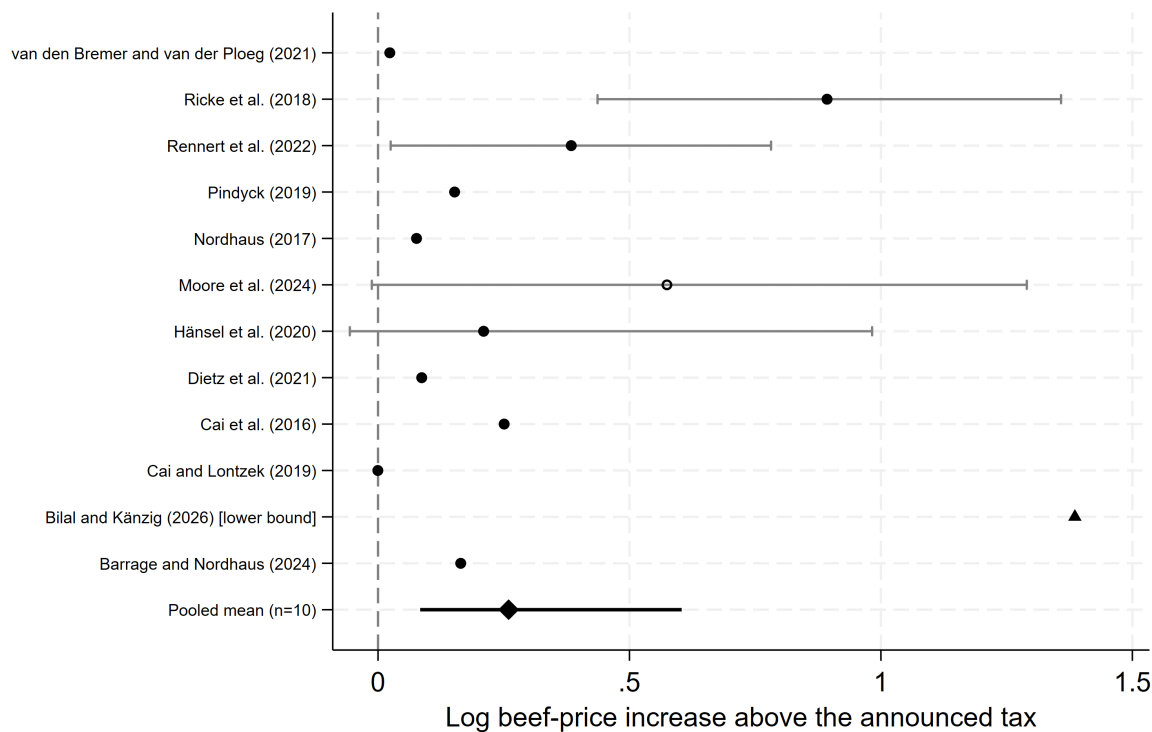


Figure 2: SCC estimates and implied beef-price gaps

Source: The studies listed in Appendix Table A1.

Notes: Each point maps a preferred SCC estimate into the log beef-price increase needed in addition to the announced Danish tax, using 59.6 kg CO₂e per kg of beef, 6.8953 DKK/USD, and the scraped pre-announcement beef price. Source ranges preserve their original meaning. Filled circles enter the equal-paper-weight pool; hollow circles and the lower-bound triangle do not. The pooled diamond and line show the hierarchical mean and its 95 percent interval.

and $q = 59.6$ kg CO₂e/kg beef. For SCC value s and announced tax $T = 300$ DKK/tCO₂, the additional log price required is

$$g(s) = \log \left[\frac{\bar{p} + (se - T)q/1000}{\bar{p}} \right]. \quad (2)$$

At the pooled mean, $g(157.96) = 0.260$, with a pooled interval of 0.084–0.604. Figure 3 compares that benchmark with all four econometric estimates. The micro DiD confidence interval ends at 0.076, just below the pooled interval’s lower endpoint. The aggregate DiD and SDiD intervals overlap its lower tail; SC does not suggest a positive response and has no conventional confidence interval because inference is placebo-based.

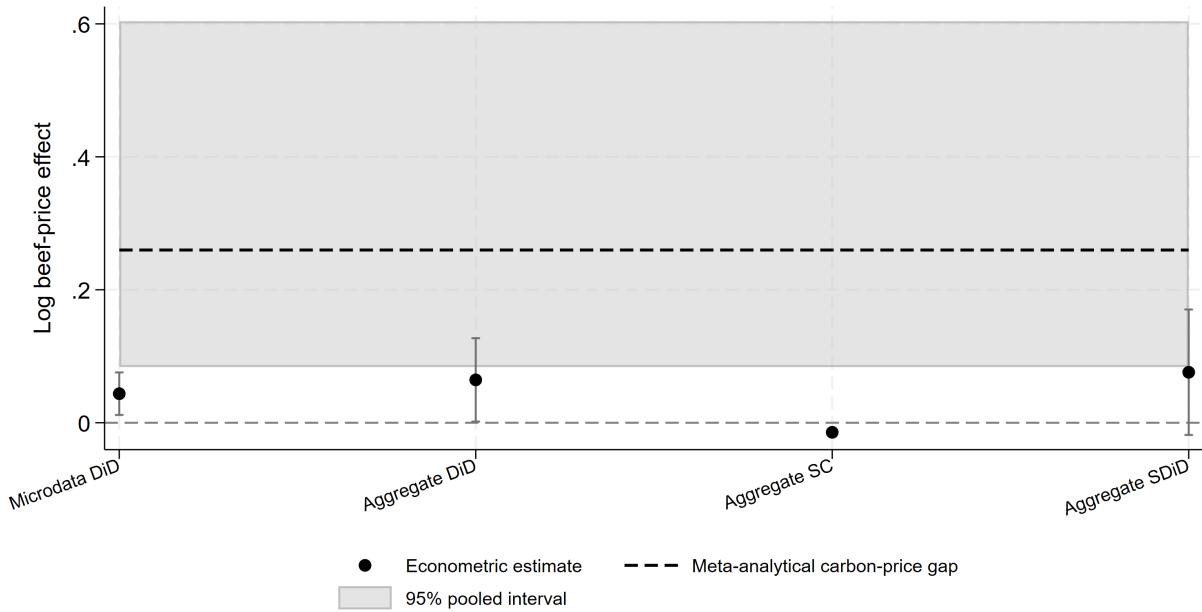


Figure 3: Econometric estimates and the meta-analytical carbon-price gap

Source: Statistics Denmark, PRIS01; dagligepriser.dk; the studies listed in Appendix Table A1.

Notes: Dots are the four ATT estimates in Table 1. Error bars are 95 percent intervals where conventional or placebo standard errors are available; SC is reported as a point with its placebo p-value in Table 1. The dashed line is the log beef-price gap implied by the pooled SCC mean, and the grey band maps its hierarchical 95 percent interval into price space.

The back-of-the-envelope magnitudes are informative but not a welfare calculation. The micro DiD point estimate corresponds to about DKK 7.1/kg at the pre-period mean, or USD 17/tCO₂. Combining this announcement-equivalent signal with the announced gross tax gives roughly USD 61/tCO₂, well below the USD 158 pooled mean. Matching the mean mechanically would require a remaining statutory component of about USD 141/tCO₂, or DKK 970/tCO₂, before deductions and incidence adjustments. This calculation assumes a common beef carbon intensity and full translation of a product-price change into a carbon-price equivalent; neither assumption is structural.

6 Discussion and Conclusion

Two independent DiD designs find a modest increase in beef prices following Denmark’s announcement. Their statistical compatibility is notable because the sources differ sharply in provenance, sampling, and aggregation. It suggests that carefully classified scraped prices can recover economically relevant aggregate movements faster and at much greater cross-sectional scale than official indices alone. That is evidence in favor of reliability and external validity for this application, not a blanket endorsement of non-official data: product coverage is selected, price normalization is noisy in levels, and product availability is intermittent.

The synthetic estimators prevent an overly tidy conclusion. Aggregate SDiD is positive but imprecise, and SC is close to zero with poor placebo support. With only 53 official series and 15 pre-treatment months, donor-weight selection is intrinsically fragile. The preferred interpretation therefore rests on triangulation: the two DiD estimates agree, their event studies do not show differential pre-trends, and the direction is supported by SDiD, while the SC discrepancy remains a material caveat.

The SCC synthesis also changes the policy conclusion. Nordhaus’s 2017 value is one informative benchmark, but it is not treated as ground truth. Including time-invariant estimates from the wider literature raises the pooled mean and, once published ranges are propagated, produces substantial uncertainty. The announcement effect is economically meaningful but does not close the central pooled carbon-price gap. Yet the lower end of the pooled range overlaps some aggregate econometric intervals, so the evidence does not identify a single corrective tax with useful precision.

Three limitations are central. First, the announcement was national and there is no untreated Denmark; identification is relative to other food prices and depends on the absence of beef-specific shocks before October 2025. Second, text rules and quantity parsing can misclassify products or create extreme normalized levels, although logs, fixed effects, explicit exclusion rules, and agreement with official CPI mitigate this concern. Third, SCC papers estimate heterogeneous objects under different discounting, damage, and uncertainty assumptions. The meta-analysis represents that heterogeneity rather than resolving it. Credible announcements can move prices before implementation, but later tax design should be updated using observed incidence, farm-level emissions, and post-implementation evidence.

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A Additional Results

Table A1: Preferred SCC benchmarks and pooling status

Study	Method	Reference year	2024 USD/tCO2	Low	High	Pool status
Cai et al. (2016)	Stochastic IAM	2015	153.5	—	—	Included
Nordhaus (2017)	DICE IAM	2030	74.2	—	—	Included
Ricke et al. (2018)	Spatial IAM	2020	599.9	254.6	1,158.0	Included
Cai–Lontzek (2019)	Stochastic IAM	Undated	43.4	—	—	Included
Pindyck (2019)	Expert elicitation	Undated	107.0	—	—	Included
Hänsel et al. (2020)	Updated DICE	2020	133.7	22.5	688.2	Included
van den Bremer–van der Ploeg (2021)	Asset-pricing IAM	2015	52.7	—	—	Included
Dietz et al. (2021)	Meta-analytic IAM	2020	78.5	—	—	Included
Rennert et al. (2022)	Modular IAM	2020	224.2	53.3	500.6	Included
Barrage–Nordhaus (2024)	DICE-2023	2030	112.5	—	—	Included
Carleton et al. (2022)	Mortality damages	2020	21.0	−30.3	65.8	Partial; external
Moore et al. (2024)	Literature synthesis	2020	343.0	38.8	1,059.3	Double-count risk; external
Bilal–Känzig (2026)	Macro empirical	2024	> 1,200	—	—	Lower bound; external

Source: The studies listed in the first column.

Notes: Values are converted to 2024 US dollars per ton of CO2. One preferred global total-SCC estimate enters per eligible paper. Values at dates other than 2030 are treated as time-invariant when no within-paper path is reported. Barrage–Nordhaus is log-interpolated to 2030. Low and high retain their source-specific interpretation. The pooled mean gives equal weight to each included paper; partial estimates, a literature synthesis, and a lower bound are external comparisons.

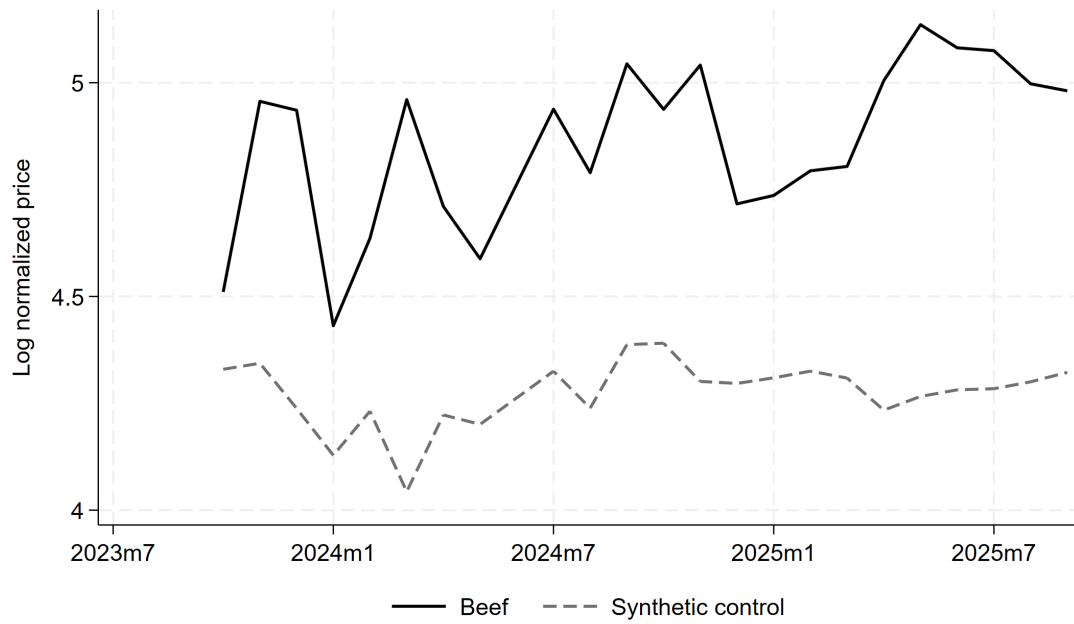


Figure A1: Scraped-data synthetic DiD: beef and synthetic control

Source: dagligepriser.dk.

Notes: Series are average log normalized prices for the treated beef commodity-store units and their Stata synthetic DiD control. The figure intentionally shows only the two series. The sample ends in September 2025.

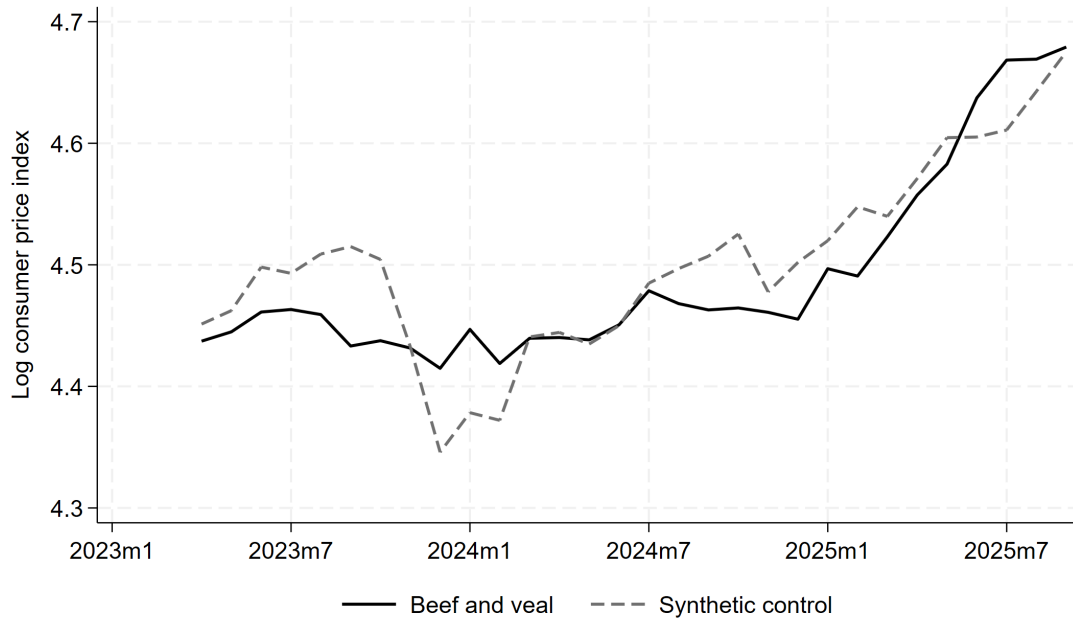


Figure A2: Official-data synthetic control: beef and synthetic control

Source: Statistics Denmark, PRIS01.

Notes: Series are log CPI for beef and veal and the Stata synthetic control over the symmetric 15-month pre/post window. The figure intentionally shows only the two series.

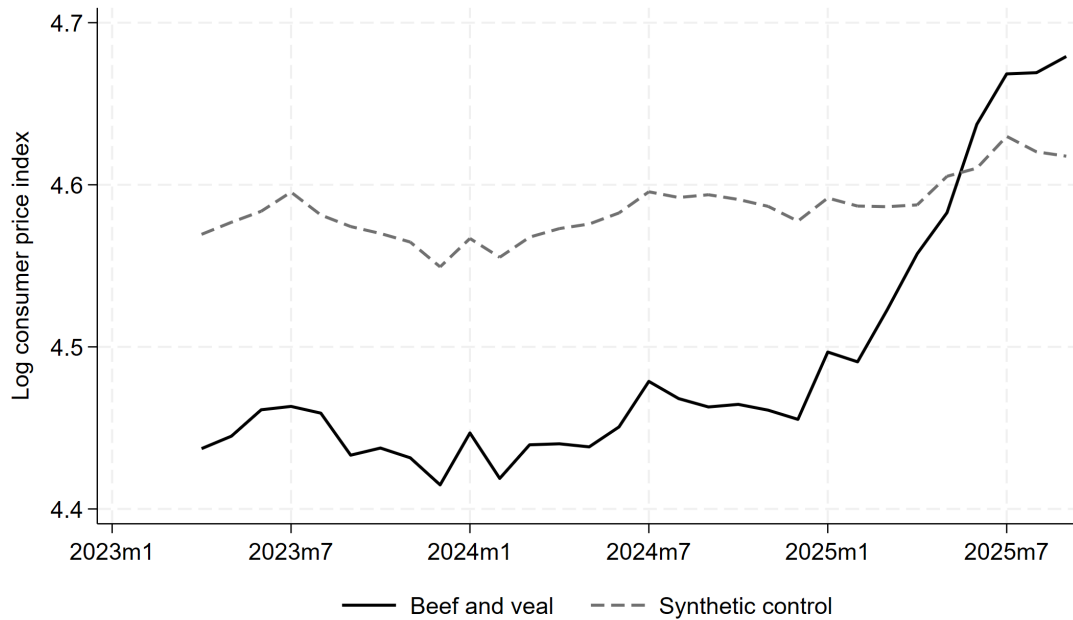


Figure A3: Official-data synthetic DiD: beef and synthetic control

Source: Statistics Denmark, PRIS01.

Notes: Series are log CPI for beef and veal and the Stata synthetic DiD control over the symmetric 15-month pre/post window. The figure intentionally shows only the two series.